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Vietnamese Legal Assistant: A Multi-agent Collaborative System using Knowledge Graph and Large Language Model

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We hope this research will be a valuable contribution to the field.

Abstract

The Vietnamese Legal Assistant System (VNLAS) is a pioneering multi-agent virtual legal assistant designed to address the lack of sophisticated legal AI systems in Vietnam. VNLAS tackles key challenges in legal Large Language Models (LLMs), including hallucinations and the accurate retrieval of legal rules, through a novel combination of enhanced datasets, model optimization, and a multi-agent framework. At its core, VNLAS utilizes a fine-tuned Qwen2.5 model, subjected to a two-stage supervised fine-tuning process leveraging Low-Rank Adaptation (LoRA), enhancing reasoning and reducing hallucinations. This approach, combined with advanced multi-task training, improves the model's ability to simultaneously handle information retrieval, legal text summarization, and case analysis. VNLAS incorporates a multi-agent architecture, mimicking roles within a legal team (e.g., researcher, editor), where specialized agents collaboratively process user queries. We enhance the retrieval process using a dynamically structured knowledge graph and a hybrid vector search. A comprehensive Vietnamese legal benchmark was developed to rigorously evaluate VNLAS, covering tasks in memorization, understanding legal concepts, and application of legal knowledge. Experimental results show that our fine-tuned embedding model achieves significant performance gains in retrieval metrics. Notably, our system demonstrates a 2.34% improvement in Mean Reciprocal Rank (MRR) and a 1.75% gain in Normalized Discounted Cumulative Gain (NDCG) for retrieval tasks and a substantial improvement in legal consultation and case analysis tasks compared to baseline and existing Vietnamese LLMs, and better performance than Gemini 1.5 Flash on many tasks. This methodology serves as a model for the development of accurate and reliable legal AI assistants in similar low-resource settings.

Keywords: Multi agent; Question Answering; Large Language Model, Knowledge Graph

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List of Abbreviation

LLMs Large Language Model.

NLP Natural Language Processing.

QA Question Answering.

RAG Retrieval Augumented Generation.

VNLAS Vietnamese Legal Assistant.

Chapter 1

Introduction

1.1 Problem statement

Legal consultation services are essential to upholding equality in society and defending human rights. Nonetheless, offering legal consultation services is severely constrained. Even though there are numerous ways to learn about the law, those who lack substantial legal knowledge and expertise may find this process to be quite challenging. Statistics from Vietnam's Ministry of Justice's Legal Aid Department show that the nation has handled 38,371 legal aid cases, of which 33,013 cases have been settled for 33,013 people (a 19% increase from 2022). Of them, 25,506 (or 77% of all instances) involved litigation (this percentage increased by 19% from 2022 to 2024)[1]. However, not all of the usage of legal consulting services has been taken into consideration. In contrast, several nations have already implemented chat bot for legal consultation, including ChatLaw[2] in China or Harvey AI[3], which helps attorneys with a range of duties. While 'AI Luat'[4] exists in Vietnam, its capacity to manage intricate legal assignments is still limited. We developed an AI-powered virtual assistant to provide legal advice and resolve user concerns in response to the growing need for legal assistance.

1.2 Motivation

In addition to scientific domains like biochemistry[5], the quick development of Large Language Model (LLMs) has spurred expansion in a number of industries in recent years, including economics (marketing[6], sales[7], advertising[8], and e-commerce[9]), education[10], healthcare[11], and finance[12]. Large language models are also anticipated to provide practical answers for legal consultation in a demanding discipline such as law[13]. Well-known models like ChatGPT[14], Gemini[15], or Llama[16], as well as models designed specifically for the legal field, can respond to user queries using the information they have been trained on and information collected from outside sources in the event that the necessary knowledge is not accessible. However, because they function based on language distribution rather than validated cases, problems like hallucination in LLMs[17] present possible concerns when utilized in the legal area. The responses they produce frequently include old or unconfirmed information, leading them to produce illusions which may be misleading or incorrect

1.3 Related Work

1.3.1 Legal Question Answering

Legal question answering is one of the most challenging tasks in Legal Artificial Intelligence (LegalAI) [18], which has been a long-standing research problem in the field of NLP generally. In recent years, some researchers have been exploring domain-specific large language models tailored across multiple scientific fields, such as biology and fintech, offering potential solutions to the challenges in legal services. Several initial attempts have been made to build legal large language models (LLMs), including chatLaw, LawyerLLama [19], and LawGPT [20]. LawGPT fine-tunes ChatGLM-6B using Chinese legal datasets and Q&A data from many competitions, while LawyerLLama uses judicial examination data to fine-tune Chinese-LLaMA-13, all of which focus on enhancing its knowledge of legal-to-dialogue scenarios.

In terms of Vietnamese legal Question Answering (Question Answering (QA)) systems, ViGPT-Law [21] is specifically fine-tuned on a legal text corpus and high-quality instruction-following language model, allowing it to generate more friendly and informative responses to law-related queries. However, the performance and reliability of Question Answering (QA) systems in the Vietnamese language are still limited, using advanced techniques such as enhanced Retrieval-Augmented Generation (RAG). Providing source pieces of information for LLMs with active retrieval improves user experience with precise answers when refining the information generated. Commercial legal AI systems like Ailuatvietnam **??** have developed strides in various legal domains, but their lack of public access reveals a significant gap in available resources for research and application. These methodologies are only used as a single LLM for legal consultations, unlike real-world law firms where staff must collaborate on legal consultations. This limitation has led to the development of ChatLaw [2] and LawLuo [22], which build a robust framework with a multi-agent system to address this issue in Chinese law datasets.

1.3.2 Multi-Agent Collaboration

In LLM-based multi-agent systems, an agent is defined as an autonomous entity capable of perceiving, thinking, learning, making decisions, and interacting with other agents [23]. Studies have proven that breaking down complex tasks into smaller subtasks and utilizing diverse roles for agents can substantially enhance the reasoning, logic, and problem-solving abilities of LLMs in general [24]. This concept could be effective for legal datasets. ChatLaw provides a multi-agent framework including legal question answering, legal analysis, and legal editor, and the results seem very optimistic, as they outperform all commercial LLMs like ChatGPT and Gemini by up to 20% in the Chinese Legal Benchmark [2].

A Retrieval-Augmented Generation (RAG) framework with LLMs can be employed as a straightforward multi-agent system, where a retrieval agent finds relevant knowledge from a vector database, and an LLM agent generates responses based on the query and retrieved knowledge [25]. The construction of knowledge graphs for Vietnamese legal cases [26] has proven effective in boosting information retrieval and analyzing cases. By connecting these entities based on their extracted relationships, these graphs create a heterogeneous representation of legal information that supports advancements in LLM systems.

Complex multi-agent collaborative systems [25] are developed to address intricate challenges across various fields. In the context of the legal system, an LLM-based multi-agent collaboration system emulates the real lawyer consultation process through role specialization and agent workflows. Standardized Operating Procedures (SOPs), modeled after actual law firm workflows, significantly reduce errors and hallucinations in legal services.

1.3.3 Legal Benchmark

The traditional methods of assessing a model's performance on a single task through fine-tuning [27], [28] are no longer sufficient for evaluating LLMs due to their rapid evolution. Lately, there has been an increasing amount of academic work devoted to creating more thorough and organized criteria for assessing different LLM capabilities. Instances include AGIEval [29], which deals with human-centric standardized tests, like those for law school admission, college entrance, maths competitions, and lawyer qualification tests; HELM [30], which, to the greatest extent feasible, measures seven metrics (accuracy, calibration, robustness, fairness, bias, toxicity, and efficiency) for each of sixteen core scenarios; KOLA [31], which concentrates on knowledge-oriented assessment under a four-level taxonomy of knowledge-related abilities; and MMBench [32], which is specifically made to assess vision-language models. There have also been works in the legal arena that resemble legal-related activities, such as Lexglue [33], which focuses on US and EU regulations, and LBOX OPEN [34], which focuses on South Korean laws. However, these works did not construct tasks in the formats that LLMs are expected to follow.

In assessing LLMs on 162 legal-related activities based on US statutes, Legal Bench was recently issued [35]. The Chinese Law-Bench [36] is used to evaluate the effectiveness of LLMs in Chinese law. In Vietnam, there is no benchmark to evaluate the effectiveness of law-related tasks for Vietnamese. Therefore, we based the method and format on the Chinese Law-Bench [36] to create a benchmark for Vietnamese law, applying Vietnamese laws and judgments crawled from the judgment aggregator site. We believe that the quality of the benchmark will be good, fully meeting the amount of knowledge needed based on Vietnamese law itself and specific situations extracted from judgments, contributing to the development of law-related applications using large language models in Vietnam.

1.4 Contributions

This study presents Vietnamese Legal Assistant (VNLAS), a multi-agent virtual legal assistant based on large language models, enhanced retrieval system to address these issues. VNLAS mitigate hallucination issues through key aspects of enhanced dataset, model optimization and multi-agent framework. Overall, our main contributions are listed as follows:

- **Dataset for evaluating performance of LLMs in legal practice:**This benchmark dataset encompasses both AI-generated and human-generated data, providing a comprehensive evaluation framework for Vietnamese Legal LLMs.
- **Multi-agent Collaboration:** LLMs System is a legal dialogue framework, VNLAS, based on the collaboration of multiple LLMs agents. Each agent is specialized in a distinct legal role (e.g., legal researcher, legal editor), enabling a workflow integration that mirrors real-world legal consultation processes.

Chapter 2

Methodology

2.1 Vietnamese Legal Assistant Framework

The Vietnamese Legal Assistant System (VNLAS) is an open-source legal LLMs that has been designed to integrate external knowledge bases its legal reasoning and decision-making capabilities. The model is built on top of Qwen 2.5 [37], which allows it to understand and generate law consultations.

One of the key innovations of VNLAS is its ability to seamlessly integrate from various curated legal databases and knowledge sources. This includes case law, regulations, and other authoritative legal materials. By combining this external knowledge with its own sophisticated language understanding, VNLAS could provide more comprehensive and accurate responses to legal queries.

We implemented an advanced Retrieval Augumented Generation (RAG) framework integrated with a dynamically structured knowledge graph to enhance legal consultation efficiency. The Legal Assistant agent identifies key legal issues, extracts relevant details from user inputs, and populates the knowledge graph. This structured representation is then utilized by the Legal Researcher agent to perform precise searches across external legal databases. By leveraging advanced Vietnamese Legal Assistant (VNLAS) techniques combined with knowledge graph reasoning, the system ensures comprehensive, accurate, and context-aware legal advice in line with current legal standards and practices. Overall, the goal of VNLAS is to provide a powerful, open-source tool that can help legal professionals, researchers, and leverage a wealth of legal knowledge. By integrating external data sources and employing advanced Natural Language Processing (NLP) techniques, the model aims to be a valuable resource for tasks such as legal research, analysis and decision-making.

2.1.1 Enhanced Datasets

This phase details the creation and enhancement of two distinct datasets: one for fine-tuning the LLMs and another for the embedding model within the RAG system.

A. Dataset for LLMs Fine-tuning

The dataset for fine-tuning LLMs consists of two primary components: a comprehensive corpus of Vietnamese legal texts and a synthetically generated dataset aimed at enhancing the LLMs' capabilities in legal reasoning and task-specific performance.

1. Real-world Legal Corpus The core dataset includes approximately 70,000 Vietnamese legal articles compiled from VLD [38]. This collection provides extensive exposure to authentic legal language, terminology, and legal structures. Additionally, over 100,000 QA-style legal advisory records were crawled from various legal consultation platforms. These records cover real-world legal queries and expert responses, ensuring contextually rich and diverse inputs for fine-tuning. Pre-processing steps included text normalization, removal of duplicates, and segmentation by legal domains to ensure relevance and consistency.

2. Synthetic Data Generation To mitigate data scarcity and enhance specific capabilities of LLMs, a synthetic dataset was generated using advanced language models such as gpt-4o-mini. This process involved creating approximately 50,000 to 60,000 instances across various legal tasks designed to improve instruction-following, formatting consistency, and legal reasoning.

Template 1: Alpaca Training Template

Below is an instruction that describes a task. Write a response that appropriately completes the request.
Instruction: {instruction} ### Response: {output}

Figure 2.1.1 demonstrates the structured prompt template employed during data generation, facilitating uniform task representation. The synthetic tasks encompassed the following key types:

- **Find Related Legal Rules:** Tasks requiring models to identify legal rules relevant to specified legal cases or queries.
- **Multiple-Choice Questions:** Approximately 30,000 multiple-choice questions were generated, focusing on core legal principles, concepts, and statutory interpretations.
- **Problem-Solving Scenarios:** Hypothetical legal scenarios were constructed, demanding the application of legal reasoning to determine relevant articles, argue legal positions, and propose solutions.

B. Dataset for Embedding Model Fine-tuning

The dataset for fine-tuning the embedding model, crucial for document retrieval in the RAG system, was constructed from three primary sources to enhance system performance. This step ensures improved retrieval accuracy by aligning vector representations of semantically similar documents, enabling precise and relevant document retrieval. By incorporating domain-specific data, the model captures nuances unique to the target application, improving relevance in specialized contexts. A well-curated dataset minimizes noise and aligns with downstream tasks, ensuring efficient and scalable performance. Additionally, leveraging diverse sources mitigates biases, resulting in a robust and balanced embedding model tailored to the RAG system's objectives. To construct the dataset, a combination of three sources was used:

1. Zalo AI Dataset The first source of data utilized for fine-tuning the embedding model in our methodology originates from the Zalo AI Challenge 2021, specifically the Legal Text Retrieval task conducted between November 15 and December 20, 2021. This dataset comprises over 60,000 legal articles spanning various domains of Vietnamese law, providing a robust foundation for training and evaluation. The training data includes legal documents paired with relevant

questions, designed to train AI models for accurate legal information retrieval. During the test phase, the dataset's original question-and-answer pairs were directly used to evaluate the model's performance, ensuring reliability in answering legal queries.

2. HUST Hackathon Dataset The second source of data for fine-tuning the embedding model is derived from the SoICT Hackathon 2024 competition, specifically the Legal Document Retrieval task. This task focuses on addressing the challenge of querying legal texts in Vietnamese. The dataset provided by the organizers includes three subsets: (1) a training dataset with 119,456 labeled query-document pairs for model training; (2) a public test set with 10,000 queries for initial evaluation; and (3) a private test set with 50,000 queries for final evaluation. All subsets share a common repository of legal documents, forming the basis for retrieval tasks. The evaluation metric used in this competition is MRR@10, which measures the system's ability to retrieve relevant documents within the top 10 results. This dataset was instrumental in fine-tuning and benchmarking our embedding retrieval model, providing a comprehensive resource for improving retrieval accuracy on legal text queries.

3. Synthetic Dataset To enhance the embedding model's performance, approximately 150,000 query-document pairs were generated using advanced language models, including GPT-4o-mini, Qwen 2.5-14B [37], Llama 3.1-8B [39], and Gemma2-9B [40]. This synthetic dataset complements real-world legal data by generating diverse, nuanced query-article pairs based on a corpus of legal texts.

Template 2: Template

Context information is below:

`{context_str}`

Based on the context, generate questions only from the provided content.

You are a Teacher/Professor. Your task is to create `{num_questions_per_chunk}` questions for a quiz/examination. Ensure question diversity across the document while staying within the context information.

Using Llama-Index with a structured prompt strategy, the models simulated a teacher creating exam questions, ensuring semantic and contextual relevance. The prompts guided models to produce legally accurate and context-aware questions linked to specific legal articles. Figure 2.1.1 illustrates the prompt format used, where context-specific prompts instructed models to generate questions focusing solely on the provided content.

The involvement of multiple models ensured broader data diversity compared to standard single-model generation. This approach expanded the dataset's variety, covering a wide range of legal topics and enabling the embedding model to generalize better, especially for less common legal scenarios. This synthetic dataset proved crucial for fine-tuning and internal evaluation, enhancing the model's robustness, contextual understanding, and retrieval performance.

The combined use of real-world legal data and synthetically generated query-document pairs addressed data limitations and enhanced the LLMs' legal information retrieval capabilities. The real-world corpus provided authentic legal language, while the synthetic dataset expanded the model's reasoning across diverse legal scenarios. This dataset underpins the supervised

fine-tuning process described next, where Low-Rank Adaptation (LoRA) is applied to integrate specialized legal knowledge efficiently.

2.1.2 Constructing and Training LLM

In this section, we present supervised fine-tuning method developed to enhance a LLMs capacity for providing accurate and contextually relevant legal consultations. The approach focuses on instruction-based fine-tuning using LoRA (Low-Rank Adaptation) to efficiently integrate specialized legal knowledge and reasoning capabilities into the model. By selectively optimizing low-rank parameters while keeping the original weights fixed, the model can deliver more informed, legally sound responses without incurring excessive computational overhead or risking catastrophic forgetting.

A. Objective To enhance the LLMs's reasoning capabilities and knowledge retrieval within the legal domain.

B. Methodology A pre-trained Qwen-7B-Instruct model was selected as the base model due to its balance of performance and efficiency. The LoRA technique [41] was applied, focusing on adapting specific layers within the model's architecture. This targeted approach minimizes the number of trainable parameters, reducing computational demands and the risk of catastrophic forgetting (where the model forgets previously learned information).

C. Mathematical Formula In LLMs such as Transformers, large weight matrices appear in the Attention mechanism and Feed-Forward layers. LoRA is integrated into these models by replacing full weight updates with low-rank matrices, reducing the number of parameters to learn and computational cost. In the Attention mechanism, the weight matrices W_Q, W_K, W_V, and W_O are used to compute query, key, value, and output vectors. With LoRA, the original weight matrices W_{Q0}, W_{K0}, W_{V0}, and W_{O0} are kept intact, and only low-rank updates are added.

The weight matrices with LoRA are expressed as:

$$W_Q = W_{Q0} + \Delta W_Q = W_{Q0} + A_Q B_Q^\top, \tag{2.1}$$
$$W_K = W_{K0} + \Delta W_K = W_{K0} + A_K B_K^\top, \tag{2.2}$$
$$W_V = W_{V0} + \Delta W_V = W_{V0} + A_V B_V^\top, \tag{2.3}$$
$$W_O = W_{O0} + \Delta W_O = W_{O0} + A_O B_O^\top, \tag{2.4}$$

where:

- $A_* \in \mathbb{R}^{d \times r}$,
- $B_* \in \mathbb{R}^{k \times r}$,
- $r \ll \min(d, k)$ is the rank of the low-rank approximation.

The query, key, and value vectors are computed as follows:

$$Q = XW_Q = X(W_{Q0} + A_Q B_Q^\top) = XW_{Q0} + XA_Q B_Q^\top, \tag{2.5}$$
$$K = XW_K = X(W_{K0} + A_K B_K^\top) = XW_{K0} + XA_K B_K^\top, \tag{2.6}$$
$$V = XW_V = X(W_{V0} + A_V B_V^\top) = XW_{V0} + XA_V B_V^\top, \tag{2.7}$$

where $X \in \mathbb{R}^{n \times d}$ is the input matrix (n is the number of tokens, d is the hidden size).

The Attention mechanism is then computed as:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V. \tag{2.8}$$

During computation, the term $XA_*B_*^\top$ can be efficiently calculated as:

1. Compute $E_* = XA_* \in \mathbb{R}^{n \times r}$.
2. Compute $E_*B_*^\top \in \mathbb{R}^{n \times k}$.

Since r is small, these computations are less costly than using full weight matrices.
In the Feed-Forward layer, the weight matrices W_1 and W_2 are also updated with LoRA:

$$W_1 = W_{10} + \Delta W_1 = W_{10} + A_1B_1^\top, \tag{2.9}$$
$$W_2 = W_{20} + \Delta W_2 = W_{20} + A_2B_2^\top. \tag{2.10}$$

The Feed-Forward layer with LoRA operates as:

$$H = \text{GELU}(XW_1) = \text{GELU}\left(X(W_{10} + A_1B_1^\top)\right), \tag{2.11}$$
$$Y = HW_2 = H(W_{20} + A_2B_2^\top). \tag{2.12}$$

where:

- $X \in \mathbb{R}^{n \times d}$: input from the Attention layer.
- $H \in \mathbb{R}^{n \times h}$: output after the activation function (h is the hidden size of the Feed-Forward layer).
- $Y \in \mathbb{R}^{n \times d}$: output of the Feed-Forward layer.

Number of parameters in the original weight matrices:

- Each original weight matrix $W_* \in \mathbb{R}^{d \times k}$ has $d \times k$ parameters.
- The total number of parameters for the weight matrices in the Attention mechanism is:

$$\text{Params}_{\text{original}} = 4 \times d \times k.$$

Number of parameters with LoRA:

- Each pair of matrices $A_* \in \mathbb{R}^{d \times r}$ and $B_* \in \mathbb{R}^{k \times r}$ has a total of $r \times (d + k)$ parameters.
- The total number of parameters for LoRA in the Attention mechanism is:

$$\text{Params}_{\text{LoRA}} = 4 \times r \times (d + k). \tag{2.13}$$

Parameter reduction ratio:

$$\text{Reduction Ratio} = \frac{\text{Params}_{\text{LoRA}}}{\text{Params}_{\text{original}}} = \frac{4 \times r \times (d + k)}{4 \times d \times k} = \frac{r(d + k)}{dk}. \tag{2.14}$$

Since $r \ll \min(d, k)$, the parameter reduction ratio is very small, allowing LoRA to significantly reduce the number of trainable parameters. Compared to conventional full-parameter fine-tuning, which involves adjusting all elements in the original large weight matrices, LoRA optimizes only the low-rank matrices A and B, while keeping the original weights fixed. This approach not only lowers computational costs and memory requirements but also mitigates the risk of catastrophic forgetting, making it particularly advantageous for adapting LLMs to specialized domains such as legal reasoning.

2.1.3 RAG System

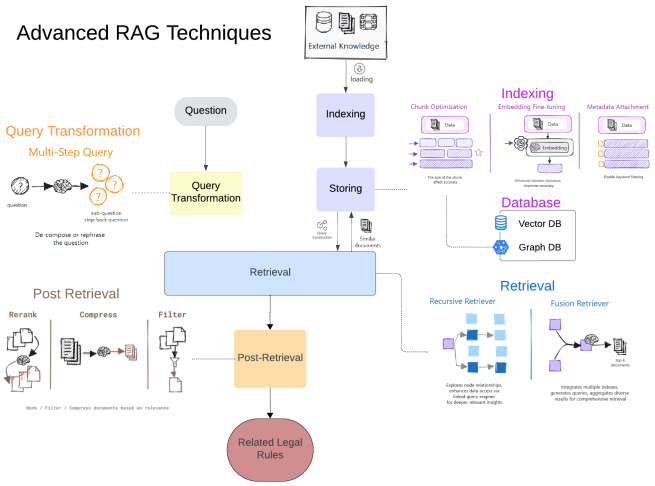


Figure 2.1: Advanced RAG for Legal Consultation

The advanced RAG system, shown in Figure 2.1, addresses challenges in legal information retrieval, such as limited reasoning capabilities, hallucinations, and context sensitivity. The pipeline begins with **indexing**, where legal documents are segmented, enriched with metadata, and embedded using fine-tuned semantic models, stored in a hybrid vector-graph database for semantic and relational queries.

Query transformation dynamically reformulates complex queries, decomposing them or enriching with contextual keywords. **Post-retrieval processing**, including re-ranking and filtering, prioritizes relevant results.

This design improves precision, mitigates hallucinations, and enhances legal reasoning by combining robust retrieval techniques with generative AI, ensuring accurate and context-aware outputs tailored to Vietnamese legal consultations.

A. Indexing and Structuring Legal Data for Retrieval

1. Data Collection and Formatting The foundation of our system begins with sourcing and structuring legal data from official repositories such as VLD (Vietnamese Legal Dictionary) [38] and other authoritative databases, such as Vietnamese Law [42]. These sources provide comprehensive legal documents, including laws, decrees, circulars, and associated legal texts.

To ensure the data is suitable for retrieval and analysis, it undergoes a multi-step formatting process:

- **Data Extraction:** Legal texts are extracted from structured and semi-structured formats

(e.g., official documents, PDF files, XML files) provided by the sources. Special attention is given to preserving the integrity of the text and its structure during this phase.

- **Normalization:** The extracted content is cleaned to remove inconsistencies such as formatting errors, redundant spaces, and irrelevant annotations. This step ensures the uniformity of legal language across the dataset.
- **Hierarchical Structuring:** Each document is organized into its hierarchical components, including titles, chapters, sections, articles, and clauses. This structuring mirrors the logical organization found in Vietnamese legal texts, enabling accurate referencing and retrieval.
- **Semantic Enrichment:** Key metadata, such as document titles, article numbers, and cross-references, is appended to the text. This metadata facilitates both semantic and relational querying in subsequent stages.

By transforming raw legal data into a structured and enriched format, our system creates a robust foundation for indexing, enabling accurate and efficient retrieval across a variety of legal formats.

2. Data Pre-processing and Indexing Building on the structured data collected from VLD [38] and Vietnamese Law [42], our system employs three key methods—chunk optimization, embedding fine-tuning, and metadata attachment to pre-process and index legal documents. These techniques ensure that the data is accurately segmented, semantically enriched, and organized for efficient retrieval.

Embedding Model Fine-tuning A crucial component of our RAG system is the embedding model, responsible for generating vector representations of legal documents. We employed the BGE-M3 model [43] for its multilingual capabilities and support for diverse text lengths. To optimize for legal information retrieval, we fine-tuned BGE-M3 using a dataset of over 400,000 query-document pairs and the SentenceTransformers library.

We employed a Multiple Negatives Ranking Loss function, which aims to maximize the similarity between a query and its corresponding relevant document while minimizing its similarity to irrelevant documents. Given a query (q) with a set of positive documents (p_i) and negative documents (n_j), the loss function is defined as:

$$\text{Loss} = \sum_{i=1}^{P} \sum_{j=1}^{N} \max\left(0, f(q, p_i) - f(q, n_j) + \text{margin}\right) \quad (2.15)$$

Where:

- P: Number of positive samples (relevant documents) for the query.
- N: Number of negative samples (irrelevant documents) for the query.
- q: The query.
- p_i: The ith positive sample (relevant document).
- n_j: The jth negative sample (irrelevant document).
- f: A similarity function (e.g., cosine similarity) between the query and the documents.

- margin: A hyperparameter defining the desired separation between positive and negative similarities.

This loss function encourages the model to learn embeddings such that the similarity between a query and its relevant documents is at least greater than the similarity between the query and any irrelevant document by a margin. This fine-tuning process, using the described loss function and our extensive dataset, significantly improved retrieval accuracy and the overall performance of the RAG system in complex multilingual legal tasks.

Chunk Optimization To improve retrieval granularity, legal documents are segmented into smaller, meaningful units (chunks). We explored various strategies, including sentence-based, paragraph-based, and fixed-size segmentation, to determine the optimal balance between context preservation and computational efficiency. The resulting chunks retain their structural hierarchy, ensuring that each unit remains contextually relevant. Ongoing research on semantic chunking further enhances the ability to capture legal content more effectively.

Metadata Attachment Metadata is systematically integrated to enrich the retrieval process by attaching essential contextual information to the indexed content. Key metadata elements include:

- **Titles:** The titles of legal documents and individual articles.
- **Content Hierarchy:** Structural information linking topics, sections, chapters, and articles.
- **Cross-References:** Relationships between legal articles, such as citations and related provisions.

The metadata is leveraged differently in each database:

- In the **vector database**, chunks of legal content are indexed alongside their associated metadata, ensuring contextually enriched semantic and lexical searches.
- In the **knowledge graph**, metadata defines hierarchical and relational structures, organizing the nodes into topics, sections, chapters, and articles. The graph also captures cross-references between articles to enable advanced relational queries.

By integrating these preprocessing and indexing techniques, our system bridges semantic and relational retrieval, ensuring comprehensive and efficient access to complex legal information.

3. Database Construction To facilitate the efficient retrieval and analysis of Vietnamese legal texts, a database architecture combining vector and knowledge graph databases has been developed. This dual-approach leverages semantic search through high-dimensional embeddings and legal reasoning through relational graph structures, ensuring precise, context-aware, and scalable information management.

Vector Database The vector database supports semantic search by indexing high-dimensional embeddings of legal clauses. Legal documents are segmented at the clause level, ensuring precise contextual retrieval. Each segment includes articles, sections, and sub-clauses, enabling fine-grained search results.

Key Attributes:

- Vectors Indexed: 563,017 dense and sparse vectors.
- Points Count: 282,569 unique legal clauses.
- Data Segments: 8 segments for optimized indexing.
- Vector Dimension: 1024 dimensions per embedding.
- Distance Metric: Cosine Similarity for relevance scoring.

Knowledge Graph Database The knowledge graph models hierarchical relationships between legal documents, supporting complex legal queries through graph traversal. It captures key legal entities and their connections, enabling logical inference and context-aware searches.

Key Attributes:

- Nodes: 85,351 total, including:
  - BoPhapDien (Root): 1 node
  - ChuDePhapDien (Legal Topics): 43 nodes
  - DeMucPhapDien (Legal Sections): 219 nodes
  - ChuongPhapDien (Chapters): 2,006 nodes
  - DieuPhapDien (Clauses): 76,658 nodes
  - TaiLieuPhapLuatGoc (Documents): 5,093 nodes
- Relationships: 251,914 edges, including:
  - BELONG_TO: 59,377 edges
  - HAS_SECTION: 59,378 edges
  - DERIVED_FROM: 56,678 edges
  - RELATE_TO: 19,803 edges
  - SOURCE_OF: 56,678 edges

Integrated System Overview The integration of vector and knowledge graph databases ensures a scalable, precise, and context-aware legal retrieval system. This structure supports semantic search through embeddings and complex legal reasoning through graph-based queries, providing comprehensive legal information management at scale.

B. Query Transformation

Query Transformation enhances retrieval effectiveness by refining user inputs through explicit rewriting and decomposition strategies, leveraging the RQ-RAG framework [44]. This process improves the alignment of queries with the indexed legal corpus, ensuring higher retrieval precision.

Query Rephrasing: Ambiguous queries are reformulated into contextually precise expressions. Using learned prompts, the model rewrites queries to better match relevant legal documents, reducing retrieval errors caused by vague or incomplete inputs.

Query Decomposition: Complex queries are decomposed into simpler, independent sub-queries. Each sub-query is processed separately, enabling granular retrieval and comprehensive aggregation of relevant legal provisions.

C. Hybrid Retrieval

Retrieval with Vector database and reranker: The retrieval component integrates vector-based and graph-based approaches, optimizing retrieval precision and context sensitivity, as illustrated in Figure 2.1. The process begins by searching the vector database using dense and sparse retrieval strategies powered by the fine-tuned BGE M3-Embedding model [43], which supports semantic retrieval across multiple granularities and languages. After initial retrieval, results are refined using the Jina-ColBERT-v2 re-ranker [45], which employs late interaction to enhance fine-grained semantic alignment between queries and retrieved documents. This combination ensures a balance between retrieval depth and efficiency.

Retrieval with knowledge graph: The fusion retriever builds upon the initial vector-based search results, identifying related legal provisions through relational insights stored in the graph database. By exploring connections between retrieved documents and their legal references, the system uncovers additional relevant content, enabling a deeper understanding of the problem and refining the scope of subsequent searches. This iterative process ensures comprehensive and context-aware retrieval for complex legal queries.

D. Post-Retrieval

Adding context for filter The system employs a two-stage process combining retrieval and re-ranking to refine search results. Initially, a knowledge graph maps relationships between legal articles, chapters, and sections within the Vietnamese legal code. This structure supports relational retrieval by connecting relevant legal provisions based on the query context, ensuring a comprehensive search across interconnected legal texts.

GPT-Based Passage Re-Ranking Following retrieval, passages are re-ranked using GPT-based reranking guided by the instructional permutation generation strategy [46]. GPT processes context-aware prompts generated from the retrieved documents, ranking passages by semantic relevance. The top-k ranked passages are then compressed into a concise representation, forming a precise input for the generative stage. This integration balances relational and semantic retrieval, ensuring context-rich and legally precise responses.

2.1.4 Implementation of the Multi-Agent Framework

The multi-agent framework for our legal information retrieval system demonstrated in figure 2.2 is designed to handle complex legal queries systematically, leveraging agents to streamline tasks and ensure accurate, contextually relevant outputs. Each agent contributes uniquely to the workflow, ensuring efficiency and precision in retrieving and processing legal information. IRAC [47] stands for four components: Issue, Rule, Analysis and Conclusion. Any situation may be broken down into these elements by applying the IRAC technique in law, which will assist your legal analysis and responses be more structured than they would be otherwise!

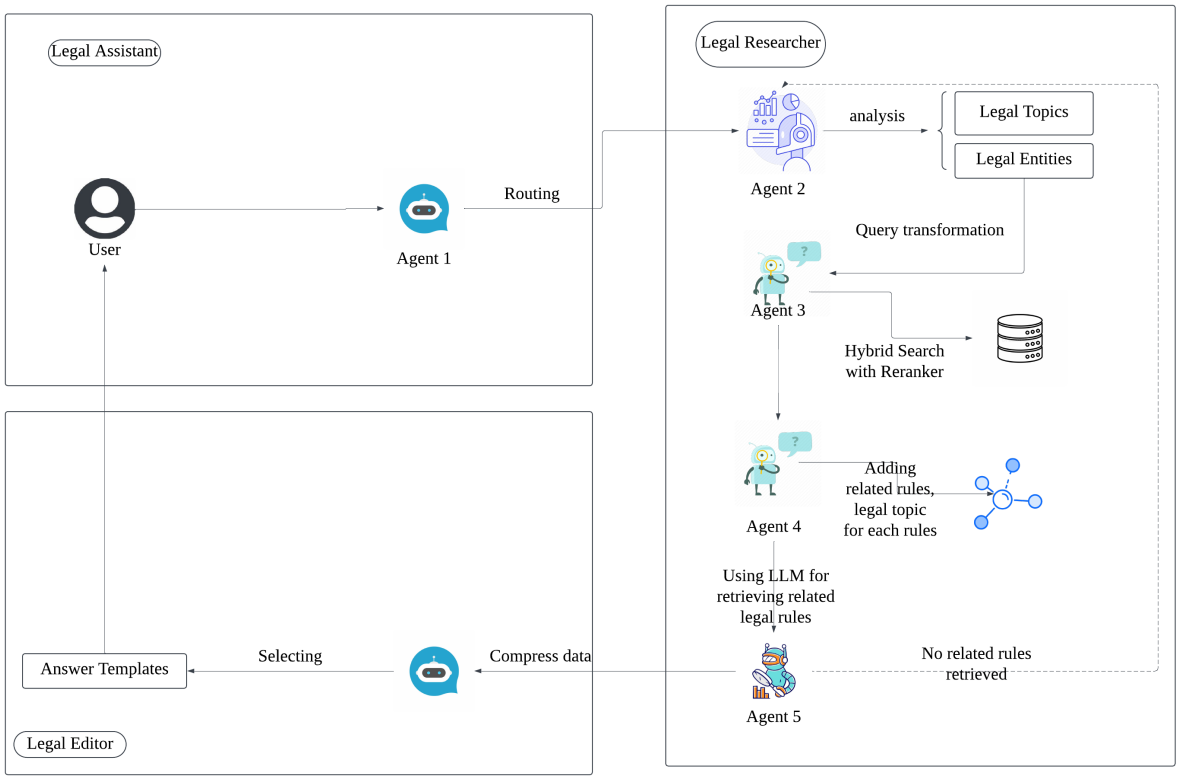


Figure 2.2: Legal Consultation Framework

A. Legal Assistant: Query Routing and Classification

The Legal Assistant serves as the entry point for user queries. Its primary responsibility is to route queries to the appropriate agents based on their type and complexity.

Agent 1: Query Router Agent 1 determines the relevance of user queries to legal issues. If the query is not related to legal matters, the system requests the user to rephrase the question with higher legal specificity. For legal queries, Agent 1 performs two critical tasks:

- **Question Classification:** Classifies queries into four categories: generation, extraction, selection, and prediction. Queries classified as extraction bypass further analysis and are sent directly to the Legal Editor for response. Other categories (generation, selection, and prediction) are flagged for deeper analysis.
- **JSON Output Generation:** Produces a structured JSON output with keys such as analysis requirement, question type, and mini-category, providing input for downstream agents.

B. Legal Researcher: Query Analysis and Retrieval

The Legal Researcher is responsible for analyzing, transforming, and retrieving legal information relevant to the user's query. This process is carried out by a series of specialized agents, each contributing a distinct capability.

Agent 2: Query Analysis and Transformation Agent 2 breaks down complex queries into manageable components, ensuring precise retrieval. Its tasks include:

- **Entity and Topic Analysis:** Identifies subjects, objects, and related legal entities within the query. Uses a dictionary sourced from Vietnamese Legal Dictionary to map the query to relevant legal topics and headings.
- **Difficulty Assessment:** Categorizes queries as easy, medium, or hard:
  - **Easy:** Basic definitions or straightforward queries involving a single law or regulation.
  - **Medium:** Requires application of multiple laws or involves simple case analysis.
  - **Hard:** Involves complex case analysis, interdisciplinary issues, or ambiguous regulations.
- **Decomposition:** Based on difficulty, queries are decomposed into sub-questions using multi-step query techniques. Easy queries are forwarded without decomposition, medium queries are divided into 2–3 sub-questions, and hard queries into up to 5 sub-questions.
- **Output Formatting:** Produces a JSON output with five keys: relevant headings, subjects, objects, legal relationship content, and decomposed questions.

Agent 3: Hybrid Search and Retrieval Agent 3 performs a hybrid search by combining dense vector search with keyword-based retrieval to ensure comprehensive results:

- Retrieves legal clauses and documents with the highest correlation to the query.
- Re-ranks retrieved documents using a pre-trained model to prioritize relevance and contextual integrity.
- Outputs the top-k results for further validation.

Agent 4: Validation and Cross-Referencing Agent 4 validates the top-k results by cross-referencing them with a knowledge graph and predefined legal datasets:

- Ensures retrieved clauses are consistent with known legal relationships and frameworks.
- Uses LLMs to assess the contextual relevance of retrieved clauses based on outputs from Agent 2. Reranks results to prioritize the most relevant provisions.

Agent 5: Recursive Retrieval and Finalization Agent 5 ensures completeness by performing the following:

- **Validation and Recursive Retrieval:** Uses LLMs to validate the retrieved legal rules. If gaps or missing provisions are detected, the system loops back to Agent 2 to initiate another cycle of analysis and retrieval.
- **Data Compression:** Compresses the final set of legal rules to extract essential information while preserving contextual integrity. This ensures concise and actionable outputs for the user.

C. Legal Editor: Answer Construction and Presentation

The Legal Editor finalizes the response to the user query by integrating the outputs from the Legal Researcher.

Agent 6: Legal Editor

- **Task Assignment:** Classifies the query's mini-category and selects the appropriate answer template from a predefined set of templates aligned with the tasks defined by Agent 1.
- **Answer Generation:** Uses the selected template to structure the response, incorporating detailed analyses, retrieved legal rules, and the user's original query context.
- **Output Delivery:** Ensures the final answer aligns with the user's query, providing clarity and actionable legal information.

The multi-agent legal framework is designed to improve **retrieval accuracy** and **reasoned thinking**, addressing two main challenges in legal AI: reducing hallucination rates and improving the precision of retrieved information. The system is structured around four primary tasks, ensuring that user queries are processed systematically and rigorously.

2.2 Dataset for evaluating performance of LLMs in Legal Practice

In this section, we have constructed a dataset to evaluate the effectiveness of LLMs for Vietnamese law, based on the LawBench [36] designed for Chinese law.

2.2.1 The Hierarchical Ability Taxonomy of The Dataset

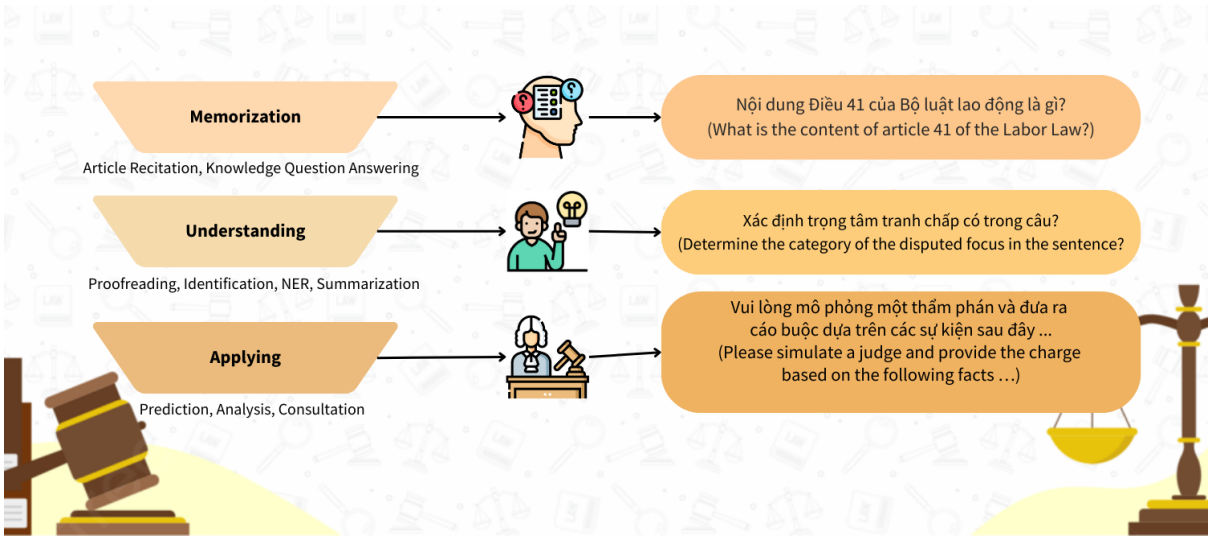


Figure 2.3: Three cognitive dimensions evaluating large language models in the dataset

It is preferred to use a range of tasks to demonstrate the capabilities of LLMs. We can gain a better understanding of the capabilities of huge language models in various areas by using a hierarchical evaluation method. We employ Bloom's cognitive model [48], which is widely used, to divide activities into various dimensions rather of just classifying them according to their difficulty. It has helped teachers create curricula and evaluate the learning results of their students. From lowest to highest, Bloom's Taxonomy categorizes cognitive learning objectives into six levels: Remember, Understand, Apply, Analyze, Evaluate, and Create. Comprehensive student growth can be encouraged by integrating learning objectives at many levels, pushing

them to advance from basic comprehension and memory to more complex analysis, evaluation, and creation. Motivated by this categorization method, we streamlined Bloom's cognitive hierarchy model [48] and retained the first three Bloom's taxonomy categories to evaluate LLMs' legal knowledge:

1.**Knowledge Memorization**: The memory level assesses the fundamental need to retain information pertaining to law. It assesses LLMs' capacity to memorize and recite fundamental legal knowledge, including rules, court decisions, concepts, common sense, legal facts, and terminology.

2.**Knowledge Understanding**: Knowing the meanings and implications of legal documents is part of the comprehension level. This includes, among other things, the capacity to understand and analyze legal concepts, documents, and issues, such as recognizing entities and relationships within legal texts and recognizing different kinds of legal issues and points of contention.

3.**Knowledge Applying**: At the application level, LLMs must combine their legal knowledge, analyze it, and respond to actual legal cases. It covers the model's numerical reasoning skills to determine the involved amount as well as its logical reasoning capabilities to provide legal consulting and judicial help.

2.2.2 Data Source and Selected Tasks

18 tasks that fit the aforementioned capability levels were chosen by us. For easier differentiation, each task is given a specific task ID. Table 2.1 contains the task list. Datasets pertaining to the same tasks may exist. When choosing datasets for each task, we select and get data from reputable sources with newest versions, combined with ChatGPT tools and examples from LawBench [36] to create QA sentences for the dataset. Furthermore, we do not currently include legal case retrieval and other tasks that require processing lengthy documents—which can exceed the length limit for most LLMs—in LawBench [36]. In order to evaluate LLMs' capacity to support legal tasks in practical situations, we have worked to design the Dataset prompts to best suit user habits and provide explicit guidance on the format of the answers.

Legal Knowledge Memorization Tasks
Legal knowledge memorization challenges investigate the degree to which legal knowledge is encoded within the constraints of LLMs. Even while external retrievers can improve this information, it is still advantageous to commit essential legal knowledge to memory. The substance of core law articles and regulations and other basic legal concepts, ideas, and principles are the two main categories of legal information that must be committed to memory. For these two categories of knowledge, we create two tasks:

1. **Article recitation (1-1):** *Given a law article number, recite the article content.* With 266 laws and legal codes, we gathered the contents of laws and regulations from the Vietnamese legal dictionary [38]. To assess the model's capacity to understand legal changes, we also included updated laws and regulations, including constitutional amendments.

2. **Knowledge question answering (1-2):** *Given a question asking about basic legal knowledge, select the correct answer from 4 candidates.* We collect the questions from Vietjack[49]. We only selected single-label questions from them in order to make it easier to find solutions during the test.

Table 2.1: Task list in Vietnamese Legal Benchmark.

Cognitive Level	ID	Task	Data Source	Metric	Type
Legal Knowledge Memorization	1-1	Article Recitation	VLD[38]	Rouge-L	Generation
	1-2	Knowledge Question Answering	Vietjack[49]	Accuracy	SLC
Legal Knowledge Understanding	2-1	Document Proofreading	VLD[38]	GLEU Score	Generation
	2-2	Dispute Focus Identification	Congbobanan[50]	F1 score	MLC
	2-3	Marital Disputes Identification	Congbobanan[50]	F1 score	MLC
	2-4	Issue Topic Identification	Congbobanan[50]	Accuracy	SLC
	2-5	Reading Comprehension	Congbobanan[50]	rc-F1	Extraction
	2-6	Named-Entity Recognition	Congbobanan[50]	soft-F1	Extraction
	2-7	Opinion Summarization	PLO[51]	Rouge-L	Generation
	2-8	Event Detection	Thuvienphapluat[52]	F1 score	MLC
	2-9	Trigger Word Extraction	Thuvienphapluat[52]	soft-F1	Extraction
Legal Knowledge Applying	3-1	Fact-based Article Prediction	Congbobanan[50]	F1 score	MLC
	3-2	Scene-based Article Prediction	Thuvienphapluat[52]	Rouge-L	Generation
	3-3	Charge Prediction	Congbobanan[50]	F1 score	MLC
	3-4	Prison Term Prediction w.o Article	Congbobanan[50]	nLog-distance	Regression
	3-5	Prison Term Prediction w. Article	Congbobanan[50]	nLog-distance	Regression
	3-6	Case Analysis	Vietjack[49]	Accuracy	SLC
	3-7	Consultation	MPS[53]	Rouge-L	Generation

Legal Knowledge Understanding Tasks

Tasks for Understanding Legal Knowledge Legal knowledge understanding tasks test the ability of LLMs to understand legal textual items, events, and relationships. Prior to applying the knowledge in practical downstream applications, it is necessary to comprehend legal texts. We chose ten challenges in total, each of which reflected varying degrees of legal knowledge and comprehension:

1. **Document Proofreading (2-1):** *Given a law article number, recite the article content.* With 266 laws and legal codes, we gathered the contents of laws and regulations from the Vietnamese legal dictionary [38]. To assess the model's capacity to understand legal changes, we also included updated laws and regulations, including constitutional amendments.

2. **Dispute Focus Identification (2-2):** *Given the original claims and responses of the plaintiff and defendant, detect the points of dispute.* The points of contention in civil lawsuits are the main points of disagreement, the intersection of inconsistencies, and the subjects that the parties are arguing about. For the advancement of the rule of law in our nation, computerized identification and detection of grounds of disagreement is both necessary and practically significant. Using ChatGPT [10], we first summarize civil judgments pertaining to disputes, then we summarize the events that transpired (question), and finally we return the sorts of disputes linked to the events mentioned (answer).

3. **Marital Disputes Identification (2-3):** *Given a sentence describing marital disputes, classify it into one of the pre-defined dispute types.* The totality of disagreements resulting from love, marriage, and divorce is referred to as marital disputes. Marital disputes are a prevalent kind of civil disputes. We synthesize from metadata the judgments crawled from Congbobanan[50] related to the topic of Marriage and Family, focusing on divorce cases, then summarize the events in those judgments (question) and point out the disputes in divorce (answer).

4. **Issue Topic Identification (2-4):** *Given a user inquiry, assign it into one of defined topics.*Legal practitioners can more effectively identify important concerns by determining the pertinent subjects to discuss during consultations. We have filtered out the verdicts and decisions related to the cases in the metadata crawled from the sentence publication page, then taken from the incident information column to make questions, then taken from the type of incident column to make answer.

5. **Reading Comprehension (2-5):** *Given a judgement document and a corresponding question, extract relevant content from it to answer the question.* Rich case details, including time, place, and character relationships, are included in judicial documents. LLMs that can intelligently read and understand court papers can help judges, attorneys, and the public get the information they need in a timely and convenient manner. We took the cases mentioned in the judgments, then exploited the basic information in the case, inspired by the information asked from the Reading Comprehension task in Lawbench [36] such as defendant information, stolen items, time of the incident.

6. **Named-Entity Recognition (2-6):** *Given a sentence from a judgement document, extract entity information corresponding to a set of pre-defined entity types such as suspect, victim or evidence* .Based on the events in the judgments, we extracted entities from each judgment, such as suspects, victims, stolen cash, property value, profits from theft, stolen property, crime tools, time, location, organization, etc. Using ChatGPT [10] with Chain of Thought method [54].

7. **Opinion Summarization (2-7):** *Given a legal-related public news report, generate a concise summary.*Important information, arguments, legal problems, applied legal principles, and the verdict are usually included in legal summaries. To increase the effectiveness of legal practitioners, it might offer a brief synopsis of the case substance. We collected news and legal news articles from Ho Chi Minh City Law Newspaper [51], then used ChatGPT [10] to summarize those news articles/news articles. We only select samples with less than 400 words to fit the length constraint of LLMs.

8. **Event Detection (2-8):** *Given a sentence from a legal judgment document, detect which events are mentioned in this sentence.* The core of facts in court cases are events. Therefore, case interpretation and other legal AI activities automatically benefit from Legal Event Detection, which is essentially significant. We used the QA Vietnamese judgments.

9. **Trigger Word Extraction (2-9):** *Given a sentence from a legal judgment document and its corresponding events, predict which words in the sentence triggered these events.* In addition to being a crucial characteristic that establishes the event category, trigger words directly cause events and offer a post-hoc explanation for the event kinds that we define. Since it is exceedingly challenging to directly identify trigger words, we made this work easier by giving the text's events along with its information, and then we looked at how well LLMs could identify event-related trigger words. We collect from the judgments in thuvienphapluat[52], then filter the keywords according to the topics in the instruction such as marriage, dispute, injury, theft. We choose small parts in the case content section related to the topic. Then we annotate by hand under the guidance of law school students.

Legal Knowledge Applying Tasks

Legal knowledge applying tasks primarily examine the ability of LLMs to not only understand legal knowledge but also simulate law professionals to apply the knowledge in solving realistic legal tasks. In the task design, we extensively examine the model's different reasoning abilities, including 3 legal content reasoning tasks: legal judgment prediction, case analysis, consultation. When predicting case judgments, judges follow a certain order when hearing a case . Therefore, in constructing the case judgment prediction task, We tackle this process by combining the SCRIBE method in prompt engineering and one-shot learning, using the ChatGPT[10], using crawled judgments from the congbobanan [50] site to generate QA benchmark sentences with the same goals and criteria as in the LawBench[36] set. We use this method to generate data for 3 tasks: fact-based article prediction (3-1), charge prediction (3-3) and prison term prediction(3-4). We further introduce the task relating to prison term prediction with article content (3-4) to examine LLMs' capability in utilizing the article content to make accurate judgment predictions. Besides, we also add the task scene-based fact prediction(3-2) to simulate judges' recognition of legal provisions .

1. **Fact-based Article Prediction (3-1):** *Given a fact statement from the legal judgment document, predict which article items should be applied.* Judges typically relate pertinent articles to the case's facts while rendering decisions. Judges can find legal articles pertaining to legal texts more rapidly with the help of article prediction. Written representations of legal norms—rules and regulations with precise definitions and legal implications—are known as legal articles. Based on the provided case description and relevant background data, the model must infer possibly applicable legal provisions. We applied the crawled verdicts from Congbobanan[50], combining the Chain of Thought method[54] and one-shot learning methods with examples taken from Lawbench[36], after being translated into Vietnamese using ChatGPT[10].

2. **Scene-based Article Prediction (3-2):** *Given a described scenario and a related question, predict the corresponding article item.* This dataset was generated by inputting legal sentences into ChatGPT[10] to construct corresponding scenario-based questions and answers. We manually selected 200 question-and-answer pairs with accurate answers from the generated dataset. Based on this, we selected 70 core legal provisions' scenario-based question-and-answer content as the test dataset.

3. **Charge Prediction (3-3):** *Given fact statement from the legal judgment document and the applied article number, predict the cause of action (charge).* The people's court formulates the cause of action, which is a synopsis of the nature of the legal relationship involved in a litigation case. Predicting the cause of action with accuracy can increase the effectiveness and equity of the legal system. A precise estimate of the cause of action can aid the court in allocating cases, allocating resources, and setting up trials during the filing and hearing of cases, increasing the efficiency and justice of the legal system. After being translated into Vietnamese using ChatGPT[10], we used the crawling judgments from Congbobanan[50], integrating the Chain of Thought method[54] and one-shot learning techniques[55] with Lawbench[36] examples.

4. **Prison Term Prediction w.o Article (3-4):** *Given fact statement from the legal judgment document, the applied article number and charge, predict the prison term..* The technique of forecasting and calculating the potential sentence a defendant would receive throughout the criminal justice process based on the case's circumstances, applicable

guiding precedents, and legal rules is known as "prison term prediction." By carefully weighing a number of variables, including the type of crime, the circumstances surrounding the incident, the social impact, and the defendant's individual circumstances, it seeks to draw reasonable conclusions regarding the length and format of the punishment. In order to replicate this process using crawled sentences from Congbobanan [50] , we combined the prompt SCRIBE approach and one-shot learning with examples from Lawbench[36] that had been translated into Vietnamese.We then supplement the provisions referred to in the judgment with content extracted from VLD [38].

5. **Prison Term Prediction w. Article (3-5):** *Given fact statement from the legal judgment document, the applied article content and charge, predict the prison term.* Retrieval-argument techniques are commonly used in LLMs to add new information. Retrieval modules that retrieve legal provisions and give comprehensive reference information for the model are also included in certain publicly available models. By combining the prompt SCRIBE approach and one-shot learning with examples from Lawbench[36] which had been translated into Vietnamese, we were able to recreate this process using crawled sentences from Congbobanan [50] using ChatGPT[10].

6. **Case Analysis (3-6):** *Given a case and a corresponding question, select the correct answer from 4 candidates.* The case analysis part tests the ability of models to analyze real cases. In order to conduct this analysis, models need to be able to reason in five different ways: word matching, concept understanding, numerical analysis, multi-paragraph reading, and multi-hop reasoning. We have selected questions from the set of questions and legal situation handling on Vietjack [49], a platform that offers a range of exercises on all types of knowledge, from exercises and questions on law to basic science knowledge like math, physics, and chemistry, in order to make the test easier to take and make it easier to learn the answers.

7. **Consultation (3-7):** *Given a user consultation, generate a suitable answer.* Legal consultation is a way for the public to access legal services. In addition to receiving support and direction, legal consultation can assist people in understanding legal challenges and in seeking specific advice and answers from qualified attorneys. To make it easier for people to get legal assistance, certain law firms and legal consulting organizations also offer online legal consultation services. We collect advice questions from the online Q and A section of the Vietnam Ministry of Public Security's information page[53] , including a variety of sections from crime prevention to traffic issues and administrative procedures.

Chapter 3

Experimentation and Results

In this section, we utilize an augmented and synthesized dataset, combined with synthetic data derived from legal codes and court rulings, to generate data for the training and evaluation of the system. The GPT-4 model is integrated to efficiently extract features for specific tasks, enhancing performance.

3.1 Datasets and Evaluation metrics

3.1.1 Data for benchmark

In particular, hallucinations pose a significant challenge not only in the legal domain but also in various applications of LLMs. To evaluate the reliability and stability of LLMs, we constructed a benchmark comprising tasks aligned with functional indicators, including:

- legal Q&A on laws and regulations,
- legal text extraction,
- case consultations,
- case analysis,
- decision-making support for predictive tasks,
- legal text inspection with tasks on spelling corrections, and
- sentence structure improvements.

To assess the performance of the constructed tasks, we employ the metrics outlined in the table 2.1, ensuring comprehensive evaluation across the defined benchmarks.

For the benchmark data for the performance index, as described in the Methodology section, we constructed it based on legal documents of Vietnam, such as the legal code, publicly available court judgments, cited legal advisories and materials published on the official government and Ministry of Public Security portals [53], among others. We have designed 18 tasks for the benchmarks, categorized into three levels: memorization, understanding, and applying. These aim to evaluate the system's effectiveness. Each task has 70 questions, which can be evaluated the performance of LLMs in various capabilities.

3.1.2 Evaluation Metrics

To measure various types of tasks, we developed a total of seven metrics:

Accuracy: Accuracy is a binary metric that evaluates whether the model prediction matches exactly the ground truth. This metric is applied to all single-label classification (SLC) tasks, including tasks 1-2, 2-4, 3-6. For SLC tasks, if multiple valid answers are extracted from the model output, the prediction is considered incorrect.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \tag{3.1}$$

F1 Score: For tasks that involve multiple output labels, the F1 score is used to calculate the harmonic mean of precision and recall. This metric is applicable to all multi-label classification tasks, such as tasks 2-2, 2-3, 2-8, 3-1, and 3-3.

$$\text{F1 Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3.2}$$

where

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3.3}$$

rc-F1: This is a specialized F1 score for the Reading Comprehension Task 2-5. It treats each token as a label, removes punctuation, stories, and extra whitespaces, performs necessary normalizations, and then computes the F1 score.

$$\text{RC-F1} = \frac{2 \cdot \mathit{Precision} \cdot \mathit{Recall}}{\mathit{Precision} + \mathit{Recall}}, \quad \text{subject to } R \geq \text{threshold}, \tag{3.4}$$

soft-F1: For extraction tasks 2-6 and 2-9, we use a soft version of the F1 score. Instead of relying on an exact match at the phrase level, this metric leverages rc-F1 for phrase matching and then calculates the F1 score on top of it. This approach is particularly useful because large language models (LLMs) often use phrasing that differs from the ground truth.

$$\text{soft-F1} = 2 \cdot \frac{\text{soft-Precision} \cdot \text{soft-Recall}}{\text{soft-Precision} + \text{soft-Recall}} \tag{3.5}$$

nLog-Distance: Tasks 3-4 and 3-5 , which involve prison term prediction, are evaluated using normalized log distance (nLog-Distance). This metric captures the continuity of prison terms by computing the logarithm of the difference between the predicted and ground truth values, normalizing the result to a range between 0 and 1 to ensure compatibility with other evaluation metrics.

$$\text{nLog-distance} = 1 - \frac{\log(|\text{Predicted Term} - \text{Ground Truth Term}| + 1)}{\log(\text{Max Term Difference} + 1)} \tag{3.6}$$

GLEU Score: For document proofreading tasks (task 2-1), the GLEU score is used. evaluates the quality of a machine-generated translation by comparing its n-grams with reference translations. It balances precision and recall by penalizing both over-translation

and under-translation. The score ranges from 0 (poor quality) to 1 (perfect match).
The mathematical formula for GLEU is:

$$\text{GLEU} = \frac{\sum_{n=1}^{N} \sum_{g \in G_n} \min(\text{count}_{\text{cand}}(g), \text{count}_{\text{ref}}(g))}{\sum_{n=1}^{N} \sum_{g \in G_n} \text{count}_{\text{cand}}(g)} \tag{3.7}$$

Where:

- N: The maximum n-gram size considered (usually 4).
- G_n: The set of n-grams of size n from the candidate translation.
- $\text{count}_{\text{cand}}(g)$: The number of occurrences of n-gram g in the candidate translation.
- $\text{count}_{\text{ref}}(g)$: The number of occurrences of n-gram g in the reference translation.

Rouge-L: Generation tasks, such as tasks 1-1, 2-7, 3-3, and 3-7, are evaluated using Rouge-L. This metric measures sentence-level structural similarity by identifying the longest matching subsequences (n-grams) between the model's output and the ground truth. It is widely used in text generation tasks.

$$\text{Rouge-L} = \frac{(1 + \beta^2) \cdot \text{Precision}_{\text{LCS}} \cdot \text{Recall}_{\text{LCS}}}{\beta^2 \cdot \text{Precision}_{\text{LCS}} + \text{Recall}_{\text{LCS}}} \tag{3.8}$$

with

$$\text{Precision}_{\text{LCS}} = \frac{\text{LCS Length}}{\text{Candidate Length}}, \quad \text{Recall}_{\text{LCS}} = \frac{\text{LCS Length}}{\text{Reference Length}} \tag{3.9}$$

and $\beta = 1$ typically assigns equal importance to precision and recall.

3.2 Experiment Settings

We trained our legal consultation Large Language Model (LLM) in two stages using an NVIDIA A10 GPU, based on the Qwen-7B-Instruct model. The fine-tuning process incorporated Low-Rank Adaptation (LoRA) , ensuring a balance between performance and computational efficiency.

Domain Adaptation with LoRA

In the first stage, LoRA was applied to adapt the model for the Vietnamese legal domain, leveraging a dataset of real-world legal consultations. This stage focused on enhancing legal reasoning and knowledge retrieval capabilities while maintaining general performance across other domains. The following hyperparameters were employed:

- **LoRA Rank:** 8
- **Target Layers:** `q_proj` and `v_proj`
- **Learning Rate:** 5×10^{-6}
- **Alpha:** 16
- **Dropout:** 0

- **Batch Size:** 1
- **Gradient Accumulation steps:** 8
- **Training Epochs:** 1

This approach allowed efficient fine-tuning on a 24GB GPU, leveraging LoRA's capability to minimize the number of trainable parameters and computational cost.

Fine-Tuning SentenceTransformers for Legal Information Retrieval

The BGE-M3 model was fine-tuned using the SentenceTransformersFinetuneEngine to adapt its embeddings for multilingual legal information retrieval tasks. This stage focused on optimizing semantic representation and retrieval performance using a dataset of over 400,000 query-document pairs. The following hyperparameters were employed:

- **Learning Rate:** 5×10^{-5}
- **Optimizer:** `adamw_torch`
- **Batch Size:** 10
- **Training Epochs:** 2
- **Warmup Steps:** 10% of total training steps
- **Weight Decay:** 0.0
- **Gradient Accumulation Steps:** 1
- **Maximum Gradient Norm:** 1.0
- **Learning Rate Scheduler:** `linear`
- **Evaluation Strategy:** `steps`
- **Logging Steps:** 500
- **Save Steps:** 500

This fine-tuning approach ensured efficient optimization of embeddings, leveraging robust loss functions and precise learning rate scheduling to adapt the pre-trained BGE-M3 model for legal domain-specific tasks.

Pipeline setup

- About the models: We use GPT4o for agent 2, finetuned LLM for other agents - About vector database: we use Qdrant for vector database - About knowledge graph: we utilize Neo4j for graph database

3.3 Results and Analysis

3.3.1 Base Embedding and Finetuned Embedding

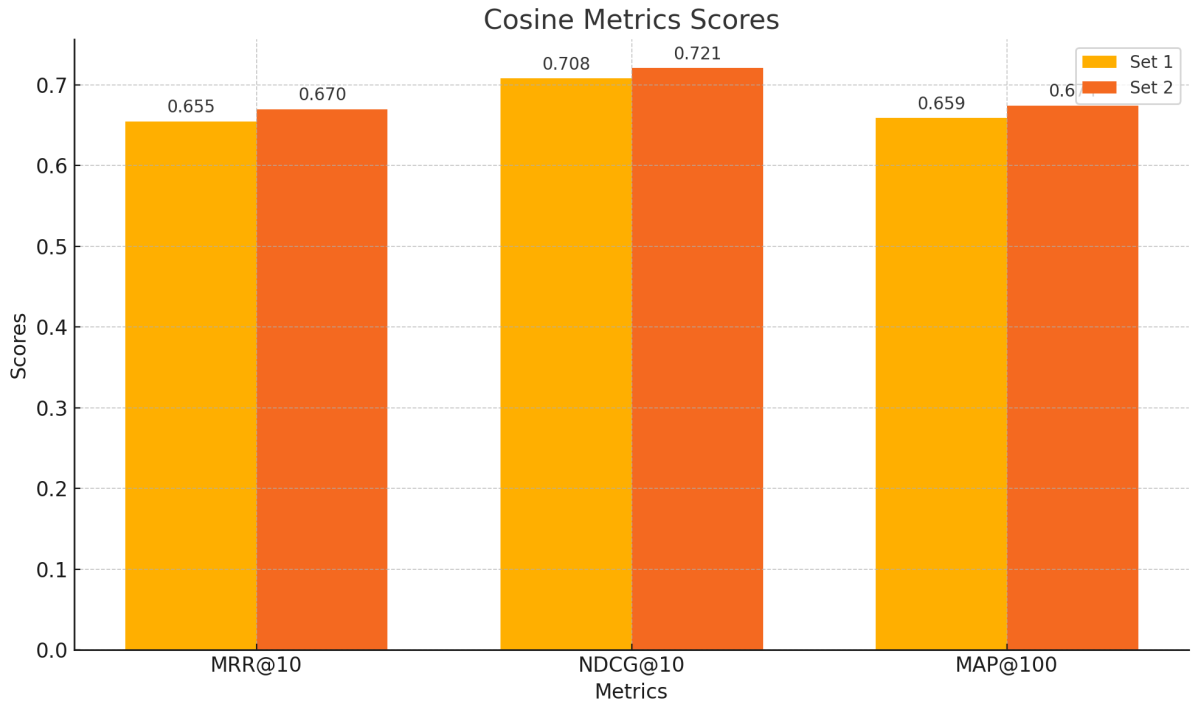


Figure 3.1: Comparison of Model Performance between the Base Model and the Finetuned Model

The evaluation metrics in Figure 3.1 compare the performance of the **base model** and the **finetuned model** using three key measures: *Cosine-MRR@10*, *Cosine-NDCG@10*, and *Cosine-MAP@100*. The results consistently demonstrate improved performance after finetuning.

Key Performance Gains

- **Cosine-MRR@10:** As shown in Figure 3.1, the mean reciprocal rank increased from 0.6549 (Base) to 0.6702 (Finetuned), reflecting an absolute improvement of +0.0153 ($\approx 2.34\%$). This indicates a higher probability of retrieving the most relevant result at the top of the ranked list.
- **Cosine-NDCG@10:** The normalized discounted cumulative gain improved from 0.7084 (Base) to 0.7208 (Finetuned), yielding a gain of +0.0124 ($\approx 1.75\%$), as depicted in Figure 3.1. This demonstrates better ranking quality within the top 10 retrieved items, ensuring more relevant results are prioritized.
- **Cosine-MAP@100:** The mean average precision rose from 0.6591 (Base) to 0.6745 (Finetuned), corresponding to an improvement of +0.0154 ($\approx 2.34\%$), as indicated in Figure 3.1. This highlights enhanced precision over a broader set of retrieved results (top 100), crucial for long-tail query performance.

The finetuned model consistently outperforms the base model across all evaluation metrics. The observed improvements in ranking precision and retrieval effectiveness suggest that finetuning

significantly enhances the model's ability to prioritize relevant results, improving its applicability in real-world information retrieval systems.

3.3.2 LLM Model base vs LLM Model fine tune

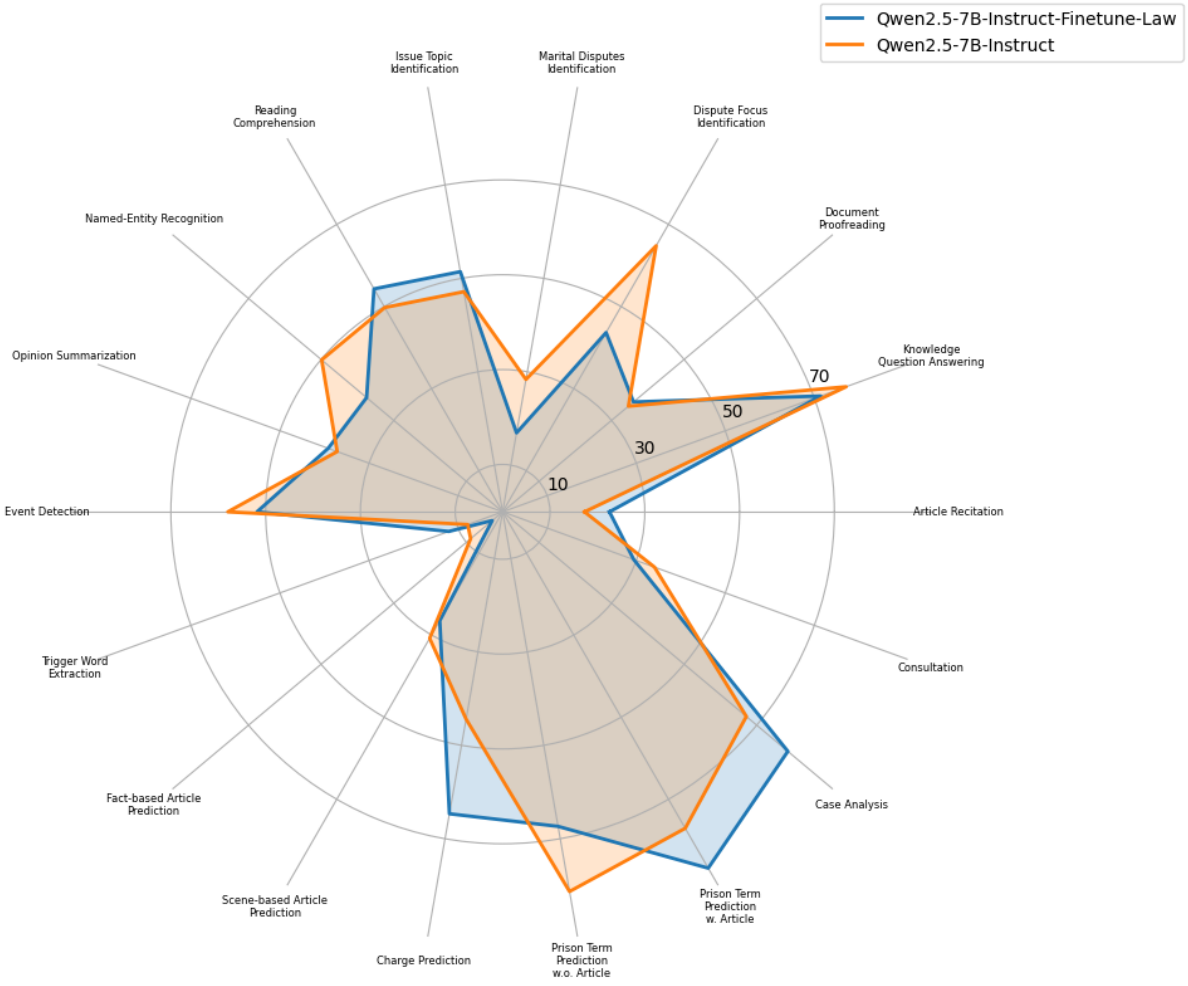


Figure 3.2: Comparison about Model Performance between model base vs model fine tune

The fine-tuned model improved in 9/18 assigned tasks, including some difficult tasks in the applying part such as charge prediction, case analysis or predicting prison sentences when provided with all relevant applicable laws as shown in the figure 3.2. This brings the potential that the model can be further developed to increase performance in handling legal-related issues.

3.3.3 Model fine tune vs other LLMs

Large language models show their strength in tasks that require general reasoning (due to the larger model size and amount of data trained). The GPT 4o model completely dominates the models in tasks that require basic understanding and logic. The fine-tuned model gives better performance than the Vietnamese-based models in tasks that require reasoning, especially case analysis, with a significant difference. However, in tasks that require understanding Vietnamese (2-6), the fine-tuned model still cannot be compared with the Vietnamese-based models.

Table 3.1: Performance of Various Models Across Tasks.

Category	**Multilingual LLMs**			**Vietnamese Oriented LLMs**		**Law-Specific LLMs**
model_name	**GPT-4.0**	**Gemini-1.5-Flash**	**Qwen2.5-7B-Instruct**	**M4uLLM-7B-Chat**	**Vistral-7B-Chat**	**Qwen2.5-7B-Instruct-Finetune-Law**
task						
1-1	10.09	17.12	17.29	25.13	25.65	22.54
1-2	87.14	74.29	77.14	14.29	30.00	71.43
2-1	65.32	49.97	34.65	31.87	37.04	36.06
2-2	24.29	64.79	64.79	59.15	35.21	43.66
2-3	38.31	32.34	28.36	12.44	0.00	16.92
2-4	81.43	72.86	47.14	42.86	0.00	51.43
2-5	59.27	58.13	49.75	59.01	56.99	54.29
2-6	60.77	51.44	49.86	42.77	32.20	37.47
2-7	43.42	41.18	37.15	40.37	44.02	39.17
2-8	70.78	57.70	57.98	25.33	19.13	51.84
2-9	28.98	31.27	7.83	11.40	7.71	12.09
3-1	96.67	21.38	8.81	74.02	88.81	2.86
3-2	42.58	21.43	30.80	35.22	34.27	26.53
3-3	72.61	70.73	44.39	52.34	63.19	64.66
3-4	86.78	61.51	81.31	86.18	88.89	67.36
3-5	85.93	82.79	77.07	76.81	79.44	86.77
3-6	90.00	78.57	67.14	7.14	47.14	78.57
3-7	38.67	34.57	34.10	36.53	35.94	29.45
Average	60.17	51.23	45.31	40.71	40.31	44.06

3.3.4 Our system vs ChatGPT 4o vs Gemini 1.5 Flash

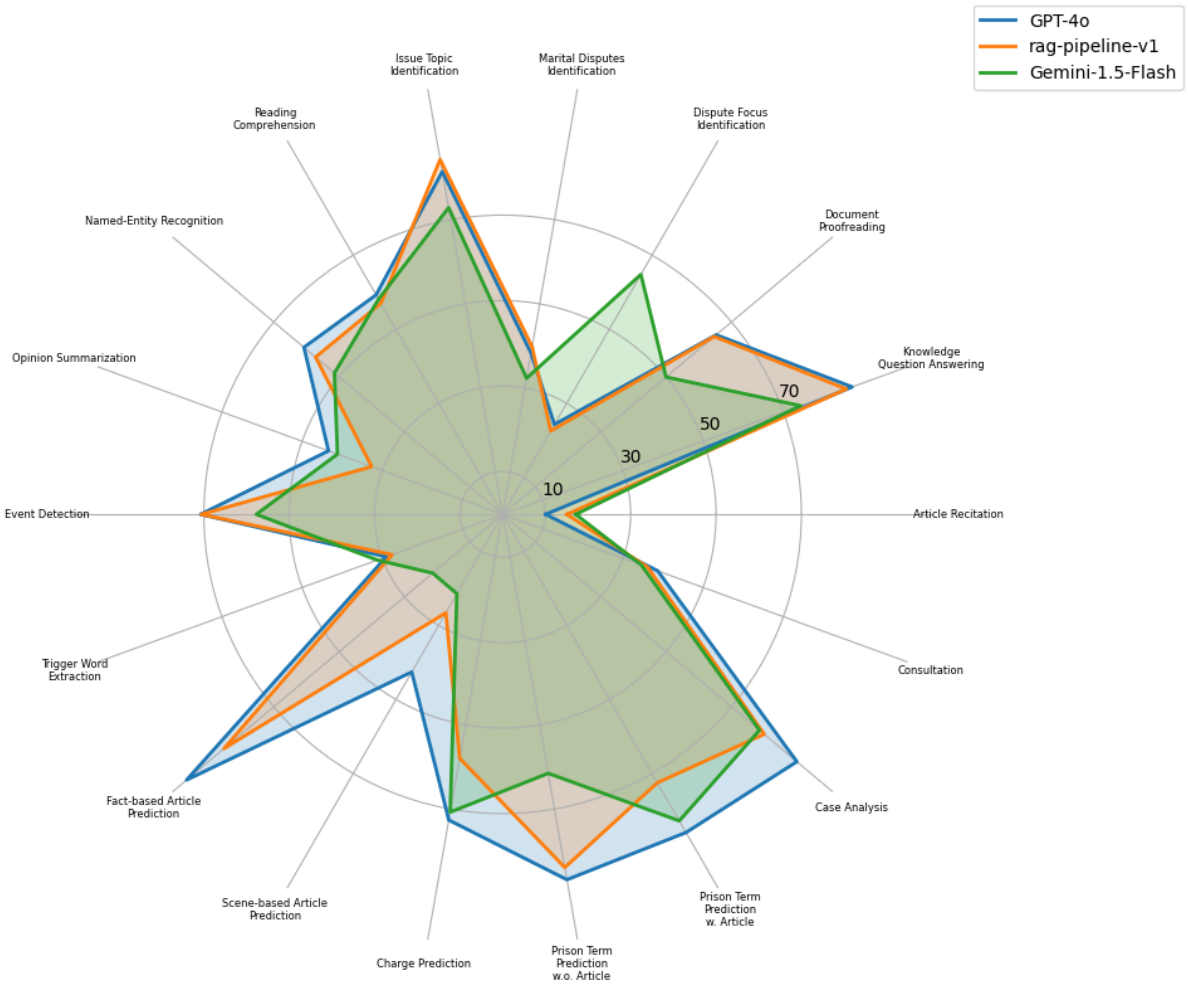


Figure 3.3: Comparison about Model Performance between model base vs model fine tune

Our system has the same score and better than gemini 1.5 flash in many tasks as shown in figure 3.3, the significant difference occurs in the tasks of fact-based article prediction, prison term prediction without article, document proofreading, event detection. Compared with gpt 4o, our system has surpassed gpt 4o in some tasks, including issue topic classification, event detection and many tasks approaching the performance of chat gpt 4o. If further improved, we believe that it has the potential to develop on par with, or even surpass, the above two models in supporting consulting on Vietnamese law issues.

Chapter 4

Conclusion and Future Work

4.1 Conclusion

In this thesis, we proposed an enhanced pipeline leveraging Retrieval-Augmented Generation (RAG) for legal consultation tasks within the Vietnamese Legal dataset. The integration of RAG enabled more effective retrieval of relevant legal information, significantly improving the system's contextual understanding and precision in answering complex legal queries.

Through fine-tuning large language models (LLMs) with domain-specific data, we achieved notable improvements in key performance metrics, including reasoning accuracy, case analysis, and legal text interpretation. The results confirmed that domain-specific fine-tuning substantially enhances the model's ability to handle specialized tasks, particularly in challenging legal scenarios involving multi-step reasoning. The incorporation of structured legal knowledge with advanced retrieval techniques led to more reliable and legally grounded responses while reducing ambiguity.

The fine-tuned embedidng model demonstrated consistent performance gains across key evaluation metrics such as *Cosine-MRR@10*, *Cosine-NDCG@10*, and *Cosine-MAP@100*, as detailed in our evaluation. The improvements highlight the effectiveness of combining symbolic reasoning with RAG-based retrieval in developing robust legal AI systems.

4.2 Future Work

Future work will focus on addressing the remaining challenges, particularly reducing hallucination effects in the generated responses. To tackle this, we plan to apply advanced fine-tuning techniques such as:

- **Direct Preference Optimization (DPO)** and **Offline Reinforcement Preference Optimization (ORPO)**: These methods will be employed to enhance the model's decision-making process by aligning its outputs with legal reasoning preferences while minimizing logical inconsistencies.
- **Temporal Optimization:** To enhance system efficiency, we aim to optimize real-time query processing through dynamic indexing, memory caching, and parallel retrieval pipelines. This will reduce response time and improve user experience in practical legal consultation applications.

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